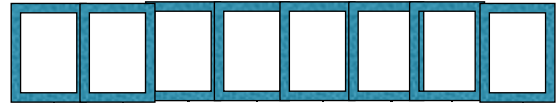


# Sample



**Q1. (2 marks) Which of the following are the properties of systematic testing procedure?**

- a. Choosing/creating test cases
- b. Black box testing
- c. Executing the test cases
- d. Generating program code
- e. Documenting the results
- f. Deciding when we are done.
- g. Analyzing requirements

**Q2. (4 points) Identify what testing activity is carried out during different SLDC phases. Write for two cases, the goal of the particular testing activity.**

| SDLC Phases:          | Testing activity:      |
|-----------------------|------------------------|
| Requirements analysis | a. Write test case     |
| Design                | b. System testing      |
| Development           | c. Integration testing |
| Integration           | d. Unit testing        |
| Maintenance           | e. Regression testing  |
|                       | f. White box testing   |

**Q3. (4 points) What is the goal of Input Robustness Testing? What are the different approaches of robustness testing? Assume we have a Stack data structure whose elements are to be reversed by a particular function. What are some Input boundary testing approaches that can be adopted here?**

**Q4. (6 points) For each of the systematic testing method given in (a), (b), (c), identify:**

(a) Functionality testing

- i. Black box or white box testing
- ii. Write completeness criteria:

(b) Input coverage testing

- i. Black box or white box testing
- ii. Write completion criteria:

(c) Hybrid of Input Partition + Shotgun Testing

- i. Black box or white box testing
- ii. Write completeness criteria:

**Q5. (7 marks)** The following is a proposed project requirement:

**Title: Inventory Management System for a Retail Store**

Our retail store urgently needs an inventory management system to keep tabs on our products, sales, and restocking activities. We require the system to manage our product inventory, track sales, and record restocking operations, all while ensuring data security. We'd also like the system to generate sales reports, alert us when products are running low, and send notifications for important events. User accounts with varying levels of access are a must, and the system should maintain comprehensive activity logs for accountability.

**Based on the above:**

**(a)** In order to run *each test case* for black-box testing the inventory management system (similar to what was expected for the Front End in Assignment #1, what **inputs** must be given to different functions (limit to 4 functions max)?

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**(b)** What expected **outputs** must be given?

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Suppose you want to introduce a new function **getNewServices** that will require you pull data from the backend database, however the database is not yet implemented. Could you still do black-box unit testing of **getNewServices**? If so, how? Briefly explain.

**Q6. (7 marks)** You are asked to write a new program that has the following requirements.

The Payout program takes as input two integers between 0 and 9.

If both numbers are the same (e.g., 4 4), then it outputs 1, except in the special case of value (9 9), when it outputs 2.

If the numbers are not equal, then it outputs 0.

The input number 0 is special: it can count as any other number (like a "wild card" in poker), so the output for the input (2 0) is 1, because there are two 2's in the input (the real one and the wild card 0).

- a. Show a set of black-box test cases, sufficient to test the implementation of this program using **output partitioning**. Use **only** as many lines as are needed.

Test output partition

Test input (values for the two integers)

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**LAST NAME:**

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**(5 marks) Mutation testing**

**(i) Is mutation testing a systematic testing method? If so, briefly describe the system for generating tests. If not, briefly explain why mutation testing is not systematic.**

**(ii) Mutation testing is a white-box method, but is different from other white-box methods such as statement coverage, condition coverage and path coverage. How is it different?**